TripSync

The World's Smartest Free Trip Planner

Product Requirements Document | International Edition | Built by Ritik
Stack: Rs.0/month until scale | Phases: 5 | Timeline: ~18 weeks

What is TripSync?

TripSync is not another travel app. It is the operating system for a trip — the single intelligent layer that replaces 10 fragmented tools with one unified workspace. Think Notion meets Google Maps meets an AI travel agent, built free, fast, and designed for the 500 million first-time travelers entering the tourism economy over the next decade.

> *"Make world-class travel planning accessible to every curious person on earth — regardless of budget, experience, or language."*

500M+	**10+**	**0**	**■0**
New travelers this decade	Apps used per trip today	Tools with constraint-aware AI	Cost to run until monetized

PART 1 — THE REAL PROBLEMS TOURISTS FACE

Problem 1 — Everything is fragmented

Flights on MakeMyTrip. Hotels on OYO. Activities on YouTube. Budget in WhatsApp. Itinerary in Notes. There is no single place where a trip lives. Tourists manually stitch 10 tools together and it always breaks.

Problem 2 — AI gives ideas, not plans

Every AI chatbot will happily list 'Top 10 places in Goa' — but none of them know that Dudhsagar Falls closes at 4pm, takes 2 hours to reach, and you also want lunch. They give you a list. You still build the plan.

Problem 3 — Group planning is chaos

6 friends planning a trip = 6 WhatsApp threads, one overworked coordinator, conflicting budgets, and someone always left out. There is no shared workspace for travel decisions.

Problem 4 — Plans die mid-trip

Internet cuts out in Spiti Valley. The hotel PDF won't open. The itinerary is in 3 screenshots. Mid-trip, travelers have zero digital support — the planning tool becomes useless on the ground.

Problem 5 — First-timers have no guide

A 21-year-old from Agra planning their first solo trip has no travel-savvy friend to call. Reddit is noise. YouTube is ads. They are making Rs.20,000 decisions with zero structured guidance.

PART 2 — HOW TRIPSYNC SOLVES EACH PROBLEM

Solution 1 — One unified trip workspace

Every element of a trip — flights, hotels, activities, routes, budget — lives in a single timeline. One link contains the entire trip. No more stitching. No more screenshots. One source of truth.

Solution 2 — Constraint-aware AI engine

TripSync's AI doesn't just suggest places. It builds a time-validated, route-optimized day plan. It knows opening hours, accounts for travel time, fits meals into the right slots. It gives you a plan that actually works.

Solution 3 — Real-time collaborative planning

Invite the group. Everyone sees the same live itinerary. Vote on stops. Comment on options. Budget split auto-calculated per person. Like Notion + Figma, but built entirely for trips.

Solution 4 — Offline-first PWA + trip mode

Install TripSync on your phone before you leave. Full itinerary, maps, and stop details cached locally. Works with zero internet. Navigate, check-in, journal — all offline, always available.

Solution 5 — An AI travel companion, not a list

The AI assistant knows your itinerary. Ask it anything mid-plan: 'What's the best time to visit this waterfall?' or 'I have 2 extra hours — what's nearby?' It answers in context, not in generic search results.

PART 3 — COMPETITOR ANALYSIS

Tool	What it does well	What it still lacks	Our edge
Wanderlog	Organizes day-by-day with maps	No constraint-aware AI, no real routing logic	AI validates timing + routes
Layla AI (YC)	Chat-first suggestions	No structured itinerary, no booking, no offline	Structured plan + offline PWA
TripIt (SAP)	Auto-builds from emails	Enterprise-focused, no AI planning	First-time traveler focus
Google Trips	Discontinued by Google	No longer exists	We fill the exact gap it left
ChatGPT	Creative suggestions	No real-world constraints, no booking	Constraint engine + full stack

The gap no one has filled: constraint-aware AI + real-time collaboration + offline-first + built specifically for emerging market first-time travelers. That is TripSync's exact position.

PART 4 — TARGET USERS

Persona	Profile	Core pain	How TripSync helps
The First-Timer	Age 18-28, tier-2/3 city, first solo trip	Zero structure, ocean of info, no guide	AI builds the full plan in 2 minutes
The Group Coordinator	Age 22-32, organizing 4-10 people	WhatsApp chaos, conflicting budgets	Real-time collaborative workspace
The Weekend Warrior	Age 25-38, urban professional	Wants a solid plan in under 5 minutes	Fastest AI itinerary generator
The Backpacker	Budget-conscious, flexible schedule	Loses access mid-trip in remote areas	Offline-first PWA, works anywhere

PART 5 — COMPLETE FEATURE SET

AI Itinerary Engine

Input: destination, dates, travel style, interests, budget, group size. Output: structured day-by-day plan with place name, category, time slot, duration, estimated cost, travel time, travel mode, and notes. Constraint engine validates opening hours, realistic travel time, meal slots, and crowd times.

Interactive Map View

All stops pinned on map, connected by route lines. Click any pin to see stop details sidebar. Toggle between Day view and Full trip view. Color-coded by day.

Smart Itinerary Editor

Drag to reorder stops within a day. Move stops between days. Add places via Google Places search. Remove or replace with AI suggestion. Inline editing of time, duration, notes.

Real-Time Group Workspace

Invite via link — no account needed to view, account needed to edit or vote. Live cursors on itinerary. Vote on stops (up/down). Comment threads on stops. Change history showing who moved what and when.

Budget Intelligence

Set total budget at trip level. Each stop has AI-filled estimated cost (editable). Running total of spent vs remaining. Per-person split calculator. Currency conversion with offline-cached rates. Budget alerts.

Offline-First PWA

Full itinerary available offline. Cached maps for booked area. Works after initial sync with zero internet. Background sync when connectivity is restored. Installable on Android and iOS.

Mid-Trip Mode

Start Trip button activates trip mode. Shows current day, next stop, time remaining. One-tap navigate to next stop. Mark as visited, skipped, or loved. AI suggests replacement if stop is skipped.

Trip Journal

Auto-generates draft journal entry from visited stops. Add photos and notes per stop. End of trip: auto-compiled journal. Export as PDF or share as public link.

AI Assistant Chat

In-context chat that knows your current itinerary. Ask anything: best time to visit, add a sunrise spot, flag non-veg stops for vegetarians, suggest nearby alternatives with extra time.

Community Discovery Feed

Public itineraries browsable by destination, duration, budget, style. One-click clone any itinerary. Each public itinerary becomes an SEO page indexed by Google — massive organic traffic channel.

PART 6 — COMPLETE FREE TECH STACK (RS.0/MONTH)

Every tool listed here has a free tier sufficient to support 5,000+ users. Paid upgrades are only needed post-monetization. For testing and development, Google Maps API is used (USD 200/month free credit = ~28,000 map loads). At scale, this can be swapped to Leaflet + OpenStreetMap at zero cost.

Layer	Tool	Purpose	Free Tier
Frontend	Next.js 14 + App Router	Framework — SSR, routing, API routes	Free forever (open source)
Frontend	Tailwind CSS + shadcn/ui	Styling + component library	Free forever
Frontend	Framer Motion	Animations and micro-interactions	Free for open source
Backend	Next.js API Routes	Serverless backend — no separate server	Free on Vercel
Database	Supabase PostgreSQL	Primary database	500MB DB free
Auth	Supabase Auth	Email + Google OAuth	50,000 MAU free
Realtime	Supabase Realtime	Live group collaboration sync	Free tier
Storage	Supabase Storage	Photo uploads for journal	1GB free
Cache	Upstash Redis	Rate limiting and caching	10k requests/day free
AI Primary	Google Gemini Flash	Itinerary generation	1,500 requests/DAY FREE
AI Backup	Groq API (LLaMA 3.1)	Fallback if Gemini rate-limited	Free tier, ultra fast
Maps	Google Maps JS API	Interactive maps (testing phase)	USD 200/month free credit
Maps (prod)	Leaflet + OpenStreetMap	Maps at scale, zero cost	Free forever
Geocoding	Google Geocoding API	Place to coordinates	Under USD 200 credit
Place Search	Google Places API	Autocomplete and place data	Under USD 200 credit
Routing	Google Directions API	Routes between stops	Under USD 200 credit
Weather	Open-Meteo API	Day-of weather for mid-trip mode	Free forever, no key
Currency	ExchangeRate API	Currency conversion	1,500 requests/month free
Email	Resend	Transactional emails	3,000 emails/month free
PDF	react-pdf	Trip journal export	Free forever (client-side)
PWA	next-pwa + Workbox	Offline caching and service workers	Free forever
Deployment	Vercel Hobby	Frontend + API deployment	Free for students
Version Control	GitHub	Code + CI/CD	Free
Analytics	Vercel Analytics	User analytics	Free on hobby tier

Layer	Tool	Purpose	Free Tier
Payments	Razorpay	Pro subscriptions	Free to integrate, 2% per txn

System Architecture

The entire application runs on Vercel as a Next.js monorepo — frontend, backend API routes, and static SEO pages all in one deployment. Supabase handles the full data layer: PostgreSQL for structured data, Auth for identity, Realtime for live group sync, and Storage for photos. AI requests go to Gemini Flash as primary with Groq as fallback. Maps use Google APIs during testing and switch to OpenStreetMap-based stack at scale.

PART 7 — DATABASE SCHEMA

Table	Key Fields	Purpose
profiles	id (FK auth.users), username, full_name, preferred_currency, language	Extended user data beyond auth
trips	id, owner_id, title, destination, lat/lng, start_date, end_date, budget_total, travel_style, interests[], group_size, is_public, share_token, status	Core trip record — one row per trip
trip_days	id, trip_id, day_number, date, day_title, notes	Each day of a trip itinerary
stops	id, day_id, trip_id, position, place_name, category, lat, lng, start_time, duration_minutes, estimated_cost, travel_time_from_prev, travel_mode, opening_hours, is_visited, is_skipped	Individual stop within a day
trip_members	trip_id, user_id, role (owner/editor/viewer), joined_at	Collaborators on a shared trip
stop_votes	stop_id, user_id, vote (+1/-1)	Group voting on individual stops
stop_comments	id, stop_id, user_id, content, created_at	Comment threads on stops
journal_entries	id, trip_id, stop_id, user_id, content, photos[], rating, created_at	Mid-trip journal with photos

PART 8 — AI PROMPTING STRATEGY

The Constraint-Aware Itinerary Prompt

The core competitive advantage of TripSync is the AI prompt that generates itineraries. Unlike generic AI travel suggestions, this prompt enforces hard real-world constraints:

- Each stop must have a realistic time slot — no teleporting, travel time between stops is accounted for
- Restaurants appear at meal times only (breakfast 7-9am, lunch 12-2pm, dinner 7-9pm)
- Attractions must be open during assigned time (museums typically close by 6pm, nightlife starts 9pm+)
- First and last day account for check-in/check-out — no stops before 2pm if arriving by flight
- Total estimated cost across all stops stays within the user's stated budget
- Maximum 5-6 stops per day — tourists get tired
- Stops are grouped geographically to minimize backtracking within a single day

The prompt returns a structured JSON object with all trip days, stops, time slots, costs, travel modes, and a trip summary. One API call to Gemini Flash generates the complete itinerary — staying well within the 1,500 free daily requests even at hundreds of users.

PART 9 — PHASED DEVELOPMENT ROADMAP

Each phase ends with something demoable and usable. No building in the dark. Every phase builds directly on the previous one — the codebase grows linearly, never requiring a rewrite.

Phase	Name	Timeline	Ships
1	Core Loop	3-4 weeks	AI itinerary generator — proof of concept
2	Visual + Editable	3-4 weeks	Map view, drag editor, public share links
3	Group + Budget	3-4 weeks	Real-time collab, voting, budget split
4	Travel Companion	3-4 weeks	PWA install, offline mode, mid-trip mode
5	Monetization	4-5 weeks	Razorpay Pro, affiliates, SEO pages, referrals

Phase 1 — The Core Loop (3-4 weeks)

Goal: Trip in, AI itinerary out. Clean, fast, impressive. Already demoable to anyone as a working product.

- Project setup: Next.js 14 + Tailwind + shadcn/ui + Supabase
- Auth: Sign up / Login / Google OAuth via Supabase Auth
- Multi-step trip creation wizard: destination autocomplete, date picker, travel style, interests, budget
- Gemini Flash API call with constraint-aware master prompt
- Itinerary display: beautiful card-based day view with category icons, time, cost, notes
- Save itinerary to PostgreSQL via Supabase
- My Trips dashboard: grid of saved trips
- Regenerate button + animated loading skeleton
- Full mobile-responsive design

Phase 2 — Make It Visual + Editable (3-4 weeks)

Goal: The itinerary lives on a map, you can edit it, you can share it. Now it feels like a real product.

- Google Maps JS API integration — all stops pinned with numbered markers, route lines, color-coded by day
- Geocoding via Google Geocoding API — lat/lng stored in DB for every stop
- Split view: left panel = day cards, right panel = map (mobile: toggle tabs)
- Drag-to-reorder stops within a day using dnd-kit library
- Move stop to different day, edit time/duration/cost/notes inline
- Add stop: Google Places search, AI auto-fills details
- AI replace: right-click stop to get AI-suggested alternative in same area
- Day summary bar: total distance, time, estimated cost
- Public share link — read-only beautiful view at /trip/[token] with 'Clone this trip' CTA

Phase 3 — Group Planning + Budget Intelligence (3-4 weeks)

Goal: Make it work for the most common real-world use case — planning with friends in real-time.

- Invite system: invite link with roles (Owner / Editor / Viewer)
- Real-time sync via Supabase Realtime — any change syncs to all open sessions instantly
- Live presence: show avatars of who is currently viewing the trip
- Voting system: thumbs up/down on each stop, sort by votes, AI re-optimization based on votes
- Comment threads on individual stops with @mention and unread indicators
- Budget tracker: running total spent vs remaining with progress bar
- Per-person cost split calculator updated in real-time
- ExchangeRate API for multi-currency support — view costs in local currency
- Activity log: right sidebar showing all changes with author and timestamp

Phase 4 — The Travel Companion — Offline + Mid-Trip (3-4 weeks)

Goal: TripSync goes from planner to companion. Works mid-trip even without internet.

- PWA setup with next-pwa — installable on Android and iOS with app icon and splash screen
- Service Worker caching via Workbox — full itinerary, map tiles, stop details cached locally
- Trip Mode UI: activated on trip start date — Today view, Map, Journal in bottom nav
- One-tap Navigate button — opens Google Maps / Apple Maps with coordinates
- Visit tracking: mark each stop as Visited, Skipped, or Loved
- AI mid-trip re-routing: skip a stop, AI suggests replacement from the same area
- Open-Meteo weather integration: day-of forecast, rain alerts, indoor alternatives suggested
- Trip Journal: add photo + note after marking visited, AI writes journal paragraph from stops
- End of trip: stats summary, auto-generated journal PDF via react-pdf, public story page

Phase 5 — Monetization + Growth Engine (4-5 weeks)

Goal: Turn users into revenue. Build the viral loop that grows TripSync without paid ads.

- Razorpay integration — Pro plan at Rs.149/month or Rs.999/year
- Free tier: 3 trips/month, 2 collaborators, no offline. Pro: unlimited everything
- Affiliate deep links on every hotel, flight, and activity stop (3-8% commission)
- 1,000+ auto-generated SEO itinerary pages via Next.js SSG — 'goa itinerary 3 days' ranks on Google
- Community feed: browse public itineraries by destination, budget, and style
- One-click clone: clone any public itinerary and edit it as your own
- Referral system: share trip, friend signs up, both get 1 month Pro free

- Email onboarding via Resend: Day 0 welcome, Day 3 tips, Day 7 upgrade prompt
- Custom analytics dashboard for DAU, itineraries generated, conversion rate, MRR

PART 10 — SUCCESS METRICS

Phase	Key Metric	Target
Phase 1	Itineraries generated	100 in first month
Phase 2	Shared trip links clicked	500 in second month
Phase 3	Group trips created	200 in third month
Phase 4	PWA installs	1,000 by end of Phase 4
Phase 5	MRR	Rs.10,000/month (67 Pro users)
12 months	Monthly Active Users	10,000+
12 months	SEO organic traffic	50,000 visits/month

PART 11 — MONETIZATION STRATEGY

Revenue Stream	Mechanism	When	Potential
Pro Subscriptions	Rs.149/month or Rs.999/year via Razorpay	Phase 5	Primary MRR
Affiliate Bookings	3-8% commission on hotel/flight/activity bookings via deep links	Phase 5	Passive, scales with users
SEO Traffic	1,000+ auto-generated itinerary pages indexed by Google	Phase 5	Free organic growth
B2B / White-label	License TripSync to travel agencies	Post Phase 5	Enterprise revenue stream

The key insight: TripSync costs Rs.0 to operate until significant scale. Gemini Flash gives 1,500 free AI generations per day. Supabase covers the database. Vercel hosts for free. You can reach thousands of users before spending a single rupee — and by then, Razorpay and affiliate revenue cover all costs.

PART 12 — WHY TRIPSYNC WINS INTERNATIONALLY

Constraint-aware AI

Not just suggestions. A plan that actually works in the real world. No competitor does this.

Offline-first architecture

Serves 4 billion people with unreliable internet. Figma-level presence for trip planning.

The SEO moat

1,000+ static itinerary pages auto-generated and indexed by Google. Compounding free traffic that no competitor can replicate without the same product foundation.

Zero cost to launch

Runs entirely on free tiers until thousands of daily active users. No VC needed. No debt. A student can build and validate this before spending a rupee.

Emerging market-first

Designed for the next 500 million travelers, not Silicon Valley users. The biggest travel market opportunity in the next decade is in India, Southeast Asia, and Latin America.

The single most important thing to understand about TripSync

It is not a travel app.

It is the operating system for a trip —

the layer that makes every other travel tool unnecessary.

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